Pranshul Pamecha

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EDUCATION

University of Wisconsin-Madison Sept 2024 – May 2028
Bachelor of Science in Computer Science | Dean's List (4x) GPA: 3.82 / 4.00
Relevant Coursework:
Data Structures & Algorithms I & II, Machine Organization & Programming, Theoretical AI, Databases, Linear Algebra, Probability

EXPERIENCE

Undergraduate Research Assistant Apr 2026 – Present
People and Robots Laboratory *Madison, WI*

- Engineered a 4-stage LLM question-generation pipeline (Gemini) converting practice conversations into a validated fill-in-the-blank question bank, enforcing **responsible-AI safeguards** (answer-leak, duplicate, and banned-answer checks) via a 3x revise-and-refilter loop.
- Designed the pipeline around dependency-injected LLM and logger interfaces for observability with an offline verification suite (3 mock-LLM harnesses, 30 invariants, 8 Jest tests), cutting full regression runs to **under a minute with 0 API cost**.
- Built an adaptive review engine (MongoDB) that ranks vocabulary by per-word error rate and exposure history, surfacing high-value review terms each session with re-queue until mastery.
- Developing the full-stack English-to-Chinese spoken-practice platform (React, Node.js/Express, MongoDB, JWT) behind an approved 50-user, 20-session study, **iterating on design through weekly syncs and thrice-weekly written updates with the research lead**, with adaptive Gemini session APIs and a 30-minute auto-submitting timed assessment.

Founder Jun 2026 – Present
QuickLearn *Madison, WI*

- Launched an adaptive SAT prep platform (Next.js, Supabase/Postgres, Cloudflare Workers, Git-based CI/CD) to **35 beta users** selected at random through social media, delivering **~11% larger weak-skill mastery gains** and routing ~30% more practice to weak skills than non-adaptive drilling.
- Shipped a real-time knowledge-tracing service (PyTorch → ONNX Runtime, FastAPI on Render) that scores per-domain mastery at **p95 14 ms** across 8 forward passes per request, with IRT 3PL and K-Means priors cold-starting students with no history.
- Built a leakage-safe evaluation pipeline over a large dataset of 101k+ simulated item attempts with chronological 70/15/15 splits, reaching **0.607 AUC against a 0.730 oracle ceiling, 83.2% of achievable**.
- Authored a **1,610-question bank** across 8 domains and 29 subskills through an LLM-assisted generation pipeline (Azure OpenAI + matplotlib figures), calibrated with IRT 3PL difficulty parameters.

Application Engineer Intern Jul 2025 – Aug 2025
Advaiya Solutions Inc. *Udaipur, India*

- Reduced end-to-end agentic AI voice assistant response latency by **58%**, enabling smoother real-time conversations through concurrent WebSocket streaming, incremental audio playback, and token-level LLM response handling.
- Co-engineered a cross-functional claim-review automation workflow **with the engineering lead and product stakeholders, iterating through daily scrums and twice-weekly progress reviews,** that reduced manual verification time by over 90% in internal testing through automated inference and structured reporting.
- Engineered repair-cost estimation logic and Approve / Flag / Reject decision rules for a vehicle claim-assessment prototype, validating estimates against historical claim bills with **94% estimate accuracy** during limited client testing.
- Trained and evaluated a YOLOv8 vehicle-damage detection prototype on 4,500 annotated images, achieving **0.86 mAP@0.5** and **0.84 precision** on a 500-image held-out test set; demoed results in a company-wide internal presentation.

PROJECTS

NutriSnap: Nutrition tracker that helps students hit macro goals from dining-hall menus **| Google Cheesehacks 2026**

- Built a **21-endpoint REST API** (Node.js, Express, Supabase) with a Gemini-vision pipeline that identifies dishes from a photo and auto-fills macros from verified menu data instead of model estimates, plus a macro-aware dining-hall recommender.
- Deployed on Render and Cloudflare Pages **for 50+ users**, cutting menu load from **1.8s** to **~0.1s (94%)** by fetching six dining halls concurrently with fault isolation, warming an in-memory cache on startup, and deduplicating items across halls.

WeTrack: Chrome extension that surfaces your own procrastination loops from browsing patterns

- Built an MV3 event-driven capture layer (service worker, alarms, idle detection) that logs activity without polling and survives service-worker termination, feeding a **zero-shot DistilBERT classifier** (FastAPI) that labels visits work vs. distraction with no per-site training data.
- Prototyped an on-device alternative to the rule-based risk scorer using a 5-8-1 PyTorch MLP trained on 3,000 synthetic sessions, exported its weights to JSON, and reimplemented inference in vanilla JavaScript **with zero runtime ML dependencies**; retained the more accurate and interpretable heuristic and built a React + TypeScript dashboard for live risk scores, browsing summaries, and historical trends.

SKILLS

- **Languages:** Java, Python, JavaScript/TypeScript, SQL, C, R
- **Frameworks:** React, Next.js, Node.js, Express.js, Flask, FastAPI, REST APIs, WebSockets
- **Tools & Concepts**: Git, Supabase/Postgres, MongoDB, Chrome Extensions API, ONNX Runtime, YOLOv8, PyTorch, Google Gemini, Azure OpenAI, GitHub Copilot, Claude Code, scikit-learn, Computer Vision, LLM Integration, Machine Learning, NLP, Responsible AI
- **Cloud & Infrastructure:** Azure, Docker, Cloudflare Pages, Render, Git-based CI/CD